



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY



Topic 2 Part 2.

Exploratory data analysis

1

1

Lecture 1 (of 2) on Exploratory Data Analysis

DEFINITIONS & SUMMARISING/VISUALISING QUALITATIVE DATA

- **Define**: statistics, descriptive statistics, inferential statistics
- **Define**: experimental unit, population, sample, statistic, parameter, variable
- **Identify**: qualitative and quantitative variables
- **Summarise qualitative variables**: class, frequency, relative frequency/proportion, percentage
- **Visualise qualitative variables**: pie and bar charts

2

2

Data Science

- Turning **data** into something meaningful
- **Science** of uncertainty
- Quintessential **interdisciplinary** science

3

3

Data Science Skillset

- Statistics, mathematics and IT skills (e.g. programming)

4

4

Data Science Skillset

- Statistics, mathematics and IT skills (e.g. programming)

5

5

Data Science Skillset

- Statistics, mathematics and IT skills (e.g. programming)
- Logical thinker
- Problem solver
- Good communicator

6

6

What does the term,

“statistics”

mean to you ?

7

7

What is / are Statistics?

A statistic*:

any quantity computed from sample data



Science of statistics:

collecting, classifying, summarizing, organizing, analyzing, estimation and interpretation of information



* Terminology also used for function to calculate the summary quantity

8

8

Role of Statistics

Field of **statistics** deals with the collection, presentation, analysis, and use of data to:

- make decisions
- solve problems
- design products and processes

It is the **science of uncertainty**

9

9

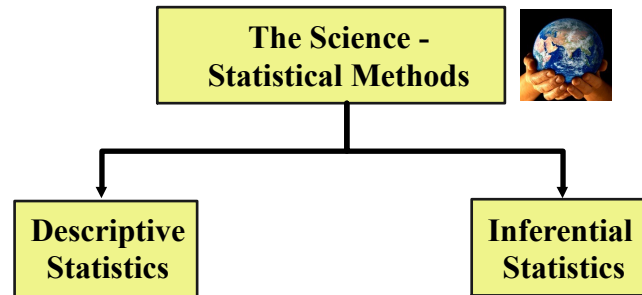
Role of Probability

- Probability provides the **framework** for the study and application of statistics

10

10

Statistical Methods



<https://maritimecyprus.com/2016/05/08/infographic-shark-attacks-2/>

11

11

Descriptive Statistics: *Science of summarizing data, numerically and graphically...*

Analysis methods applicable depends on the variable being measured and the research questions that you are trying to answer ...

12

12

Inferential Statistics: science of using the *information in your sample* to say (i.e. to “*infer*”) something *about the population of interest*

13

13

Thinking Challenge

Suppose the student newspaper is interested in what proportion of University of Galway students pay rent and the average amount of rent paid

How would you find out?



Break down the question...

14

What is the individual / **experimental unit**?

What is the **population** of interest?

What are the **variables** of interest?

What **types** are these variables?

What are the **parameters** of interest?

How would you **collect the data**?

What are the **observations** for the variables?

How would you **summarise** these observations?

15

15

Some important terms:

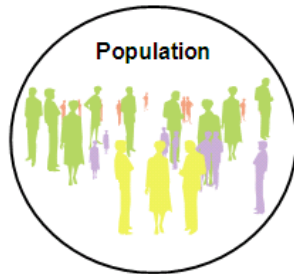


An ***experimental unit / individual***
is a single object upon which we collect data,
e.g. person,
thing,
transaction,
event.

16

16

Some important terms:



A **population** is a collection of experimental units/individuals that we are interested in studying.
e.g. people,
things,
transactions,
events

17

17

Some important terms:



18

18

Some important terms:



A sample

is a subset of experimental units / individuals from the population.

e.g. people,
things,
transactions,
events

19

19

Some important terms:

A **variable** is a quantity (eg characteristic or property) that can take different values. An individual variable **varies** from one individual (or experimental unit) to the next.

examples:



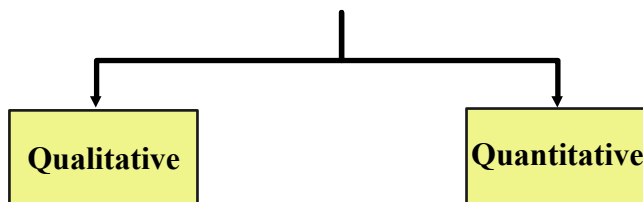
height
grade score
account balance
gender (m/f/non-binary),
letter grade (A, B, C, etc.),
Likert scale (agree, neutral, disagree, etc.)

20

20

Types of variable:

A **variable** is a quantity that can vary from one individual to the next.



May be measured, or more generally “observed”, on each individual

21

21

Qualitative Data:

Classified into categories, can be **ordered**:

- **Grade** achieved in ST2001 (A, B, C...)



or **unordered**:

- **Gender** of each employee at a company



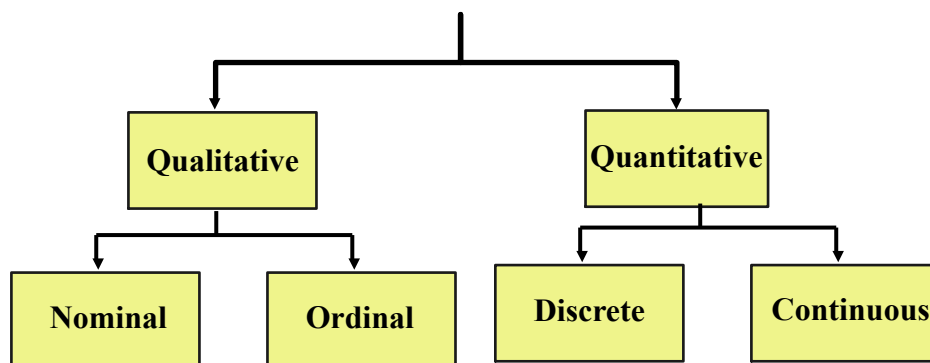
- **Method of payment** (cash, cheque, credit card)



22

Types of variable:

A **variable** is a quantity that can vary from one individual to the next.

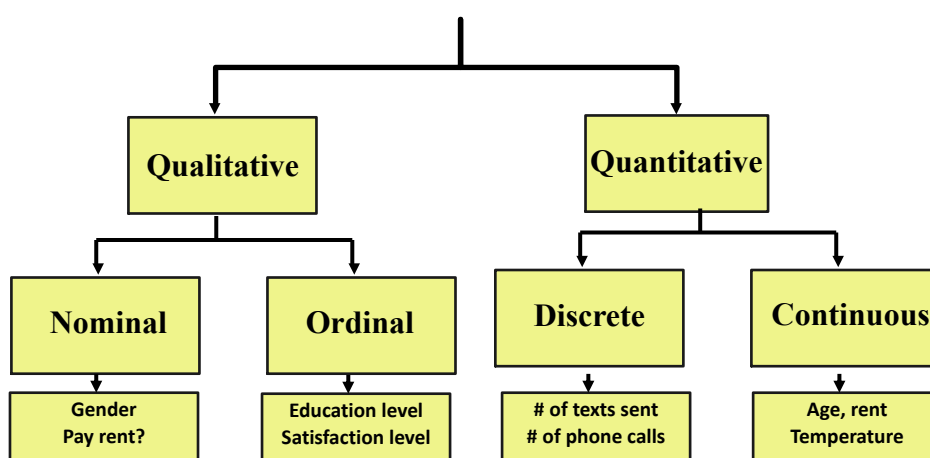


23

23

Types of variable:

A **variable** is a quantity that can vary from one individual to the next.



24

24

Gapminder Data: <https://www.gapminder.org/>

The Gapminder Foundation is a Swedish NGO which promotes sustainable global development by increased use and understanding of statistics about social, economic and environmental development

Gapminder Test

Welcome to the Gapminder Global Facts test!

You will get 13 fact questions. There's a time limit of 45 seconds per question.

If you pass the test, you are qualified to become a Gapminder and we'd like to honor you with the Gapminder Global Facts Certificate!

If you don't pass the test, don't worry: we won't tell anyone and you can try again later.

Thanks for spreading a fact-based worldview, starting with yourself.

Good luck!
The Gapminder Team

Next

0%

<http://upgrader.gapminder.org>

25

25

Summarising the gapminder data

- first install (once only), then load the `gapminder` package

```
> library(gapminder)
```
- In this course we use **packages** `dplyr` for "data wrangling" and `ggplot2` for **plotting**
- We can use base R for "data wrangling" and plotting, but these packages have some advantages
- Both packages in the `tidyverse` suite - could just load `tidyverse`

```
> library(dplyr)
> library(ggplot2)
```
- `dplyr` uses the **pipe operator** `%>%` for smooth & efficient workflows
- If you ever see an **error message** that R cannot find `%>%` it usually means you have forgotten to load `dplyr`

26

26

Gapminder Data

```
gapminder %>% head()
## # A tibble: 6 x 6
##   country    continent  year lifeExp    pop gdpPercap
##   <fct>      <fct>    <int>  <dbl>  <int>   <dbl>
## 1 Afghanistan Asia      1952   28.8  8425333  779.
## 2 Afghanistan Asia      1957   30.3  9240934  821.
## 3 Afghanistan Asia      1962   32.0 10267083  853.
## 4 Afghanistan Asia      1967   34.0 11537966  836.
## 5 Afghanistan Asia      1972   36.1 13079460  740.
## 6 Afghanistan Asia      1977   38.4 14880372  786.

gapminder %>% dim()
## [1] 1704    6
```

27

27

- What is the *typical observation*?
- Is there much *variation/spread* between individuals in the dataset?
- How are the observations distributed over all individuals in the group – i.e. what is the *shape* or *distribution*?
- Are there any values lying outside of the range where the majority of the dataset values lie – *outliers*?

Summarising data (variables) can be done **numerically**, with appropriate summaries, or **graphically**, with appropriate plots

28

28

Numerical summary of categorical data: frequency (count) and proportion

Continent	Frequency	Proportion
Africa	624	0.36619718
Americas	300	0.17605634
Asia	396	0.23239437
Europe	360	0.21126761
Oceania	24	0.01408451

```
gapminder %>% select(continent) %>% table()
## .
## Africa Americas Asia Europe Oceania
## 624 300 396 360 24
```

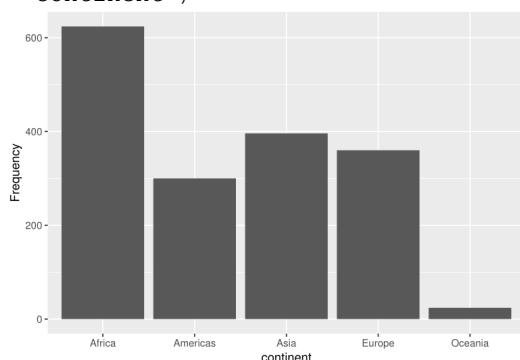
```
gapminder %>% select(continent) %>% table() %>% prop.table()
## .
## Africa Americas Asia Europe Oceania
## 0.36619718 0.17605634 0.23239437 0.21126761 0.01408451
```

29

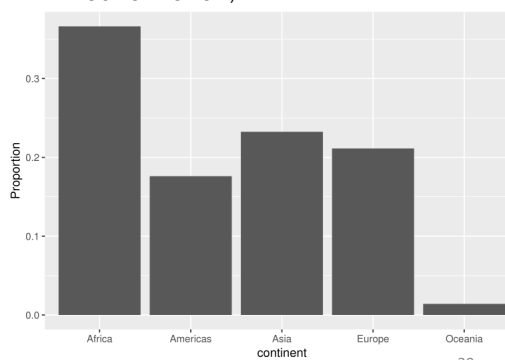
29

Graphical summary of categorical data: bar chart, pie chart

```
gapminder %>%
  ggplot(aes(x = continent)) +
  geom_bar()
labs(y = "Frequency", x =
"continent")
```



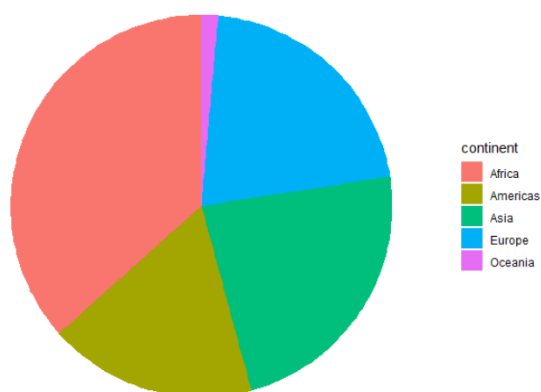
```
gapminder %>%
  ggplot(aes(x = continent, y =
after_stat(count/sum(count)))) +
  geom_bar() +
  labs(y = "Proportion", x =
"continent")
```



30

30

Graphical summary of categorical data: bar chart, pie chart



Advice: don't use pie charts

People find determining angles very difficult

Easier to understand lengths/heights

31

31

EDA Lecture 1 summary

- Terminology and definitions
- Types of data
- Gapminder data
- R packages for data wrangling and plotting
- Summarising categorical data: tables and bar charts using R

35

35

Some useful operators and functions in base R

`$` to extract a named element, by name, from an object such as a dataframe

`head()` input an R object, output print the first few elements (default 6 elements)

`dim()` input R object, output dimensions; if input is a dataframe output is number of rows and number of columns

`table()` input a variable (eg vector or column in dataframe), output table of frequencies

`prop.table()` input table of frequencies, output table of proportions

36

36

Useful functions in `dplyr`

`filter()` to filter rows (observations)

`select()` to select columns (variables)

`mutate()` to create new variables

`summarise()` `group_by()` to summarise variables

`arrange()` to sort data

37

37

LECTURE 2 (of 2) on Exploratory Data Analysis

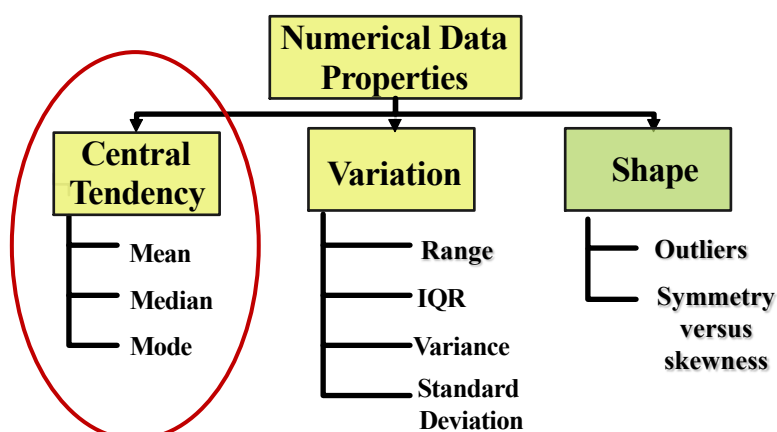
SUMMARISING AND VISUALISING QUANTITATIVE DATA

- **Summarise quantitative variables**: calculate mean, median, mode, range, quartiles, IQR, variance, standard deviation
- **Discuss** pros and cons of these numerical data summaries
- **Visualise quantitative variables**: construct & interpret histograms, boxplots
- **Use visualisations** to explore shape of distribution
- **Identify and discuss** existence of outliers

38

38

Summarising Continuous Data



40

40

Numerical summary of typical value:

Definition

Suppose that the observations in a sample are $x_1, x_2, \dots, x_n$. The **sample mean**, denoted by $\bar{x}$, is

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Sensitive to extreme values

Given that the observations in a sample are $x_1, x_2, \dots, x_n$, arranged in **increasing order** of magnitude, the sample median is

$$\tilde{x} = \begin{cases} x_{(n+1)/2}, & \text{if } n \text{ is odd,} \\ \frac{1}{2}(x_{n/2} + x_{n/2+1}), & \text{if } n \text{ is even.} \end{cases}$$

NOT Sensitive to extreme values

Mode is the most frequent observation in a dataset.

1, 1, 1, 1, 2, 1

41

41

Example

Data: breaking strength of wire in kilograms

220 214 222 218 223 210 223 210 227 225 212

- Find the median:
 - Order the data from lowest to highest

210 210 212 214 218 220 222 223 223 225 227

Median

$$\frac{11+1}{2} = 6$$

- Find the Mean:

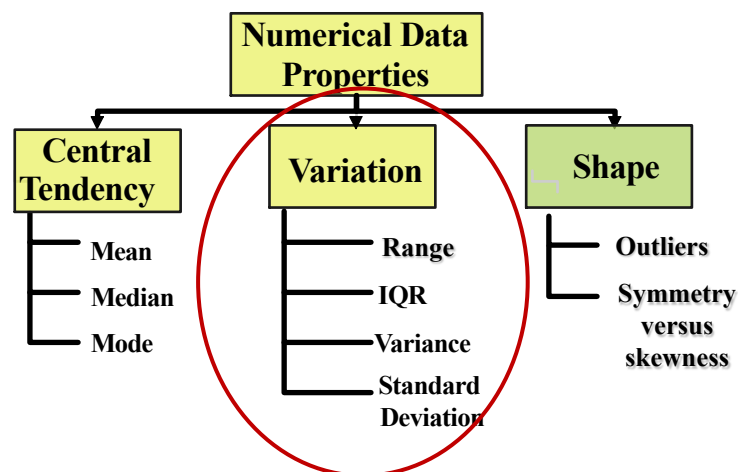
$$\text{Mean} = \frac{220 + 214 + \dots + 222}{11} = 218.5455$$

- Mode is 210 and 223, as both have been repeated twice

42

42

Summarising Continuous Data



43

43

Numerical Summary of Spread

- Range = **maximum - minimum**

Examples:

- 1, 2, 5, 8, 10 gives range of $10 - 1 = 9$
- 1, 5, 5, 5, 10 also gives range of 9
- Clearly the range is poor measure of spread
- Also badly affected by outliers

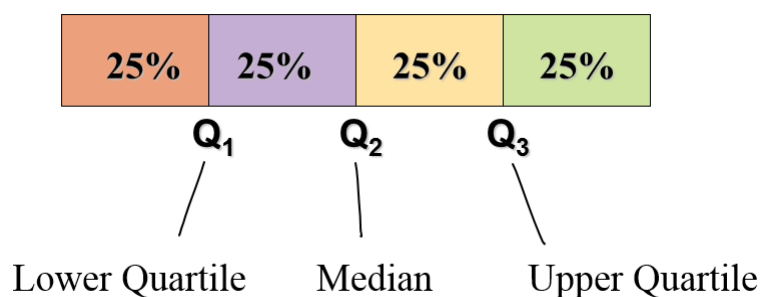
44

44

Numerical Summary of Spread

- Interquartile range ($IQR = Q_3 - Q_1$)
- Middle 50% range of data, so is robust to outliers

Split ordered data into 4 quarters



45

45

Tukey's Method for IQR (lots of others)

Data: breaking strength of wire in kilograms

220 214 222 218 223 210 223 210 227 225 212

Put data in ascending order:

210 210 212 214 218 220 222 223 223 225 227

↑ ↑ ↑

$Q_1 = 213$ Median $Q_3 = 223$

Lower (Upper) quartile is median of lower (upper) 50% of data including the median

$$IQR = Q_3 - Q_1 = 223 - 213 = 10$$

46

46

Numerical Summary of Spread

- Common measure of spread is the **standard deviation**, which takes into account how far *each* data value is from the mean
- A **deviation** is the distance of a datapoint from the mean
- Since the sum of all the deviations would be zero, we square each deviation and find an average (of sorts) of them (called the **variance**)
- We take the square-root this average squared deviation... **Why?**

Definition

The **sample variance**, denoted by s^2 , is given by

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}.$$

The **sample standard deviation**, denoted by s , is the positive square root of s^2 , that is,

$$s = \sqrt{s^2}.$$

47

47

Sample Standard Deviation

- In same units as original variable
 - So preferable to sample variance, which is in squared units
- But... it is sensitive to outliers

48

48

Example

Data: breaking strength of wire in kilograms

220 214 222 218 223 210 223 210 227 225 212

- Find the sample variance
- Find the sample standard deviation

$$\bar{x} = 218.5455$$

$$\text{Sample Variance} = s^2 = \frac{(220 - 218.5455)^2 + (214 - 218.5455)^2 + \dots + (222 - 218.5455)^2}{11 - 1} = 37.67273$$

$$\text{Sample Standard deviation} = s = \sqrt{\text{Sample Variance}} = \sqrt{37.67273} = 6.1378$$

49

49

Numerical Summary in R: Vector

```
wire.strength <- c(220, 214, 222, 218, 223, 210, 223, 210, 227, 225, 212)
```

```
> mean(wire.strength)
[1] 218.5455
> median(wire.strength)
[1] 220
> var(wire.strength)
[1] 37.67273
> sd(wire.strength)
[1] 6.137811
```

fivenum() function uses Tukey's method for Q_1 and Q_3

```
> fivenum(wire.strength)
[1] 210 213 220 223 227
```

summary() function uses a different formula for quartiles

```
> summary(wire.strength)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 210.0   213.0   220.0   218.5   223.0   227.0
```

50

50

Numerical Summary in R:

Calculate the **mean** life expectancy for `gapminder` data:

```
> gapminder %>% summarise(mean.life = mean(lifeExp))
# A tibble: 1 x 1
  mean.life
  <dbl>
1      59.5
```

Calculate the **mean** life expectancy for different continents:

```
gapminder %>%
  group_by(continent) %>%
  summarise(mean.lifeexp = mean(lifeExp))
# A tibble: 5 x 2
  continent mean.lifeexp
  <fct>         <dbl>
1 Africa         48.9
2 Americas       64.7
3 Asia           60.1
4 Europe         71.9
5 Oceania        74.3
```

arrange

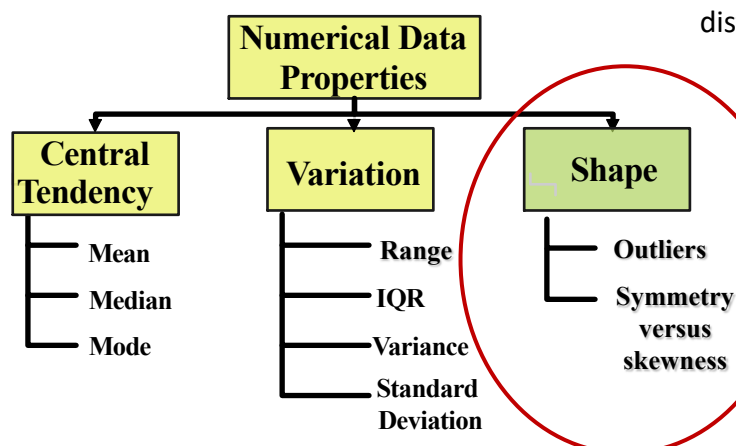
```
gapminder %>%
  group_by(continent) %>%
  summarise(mean.life = mean(lifeExp)) %>%
  arrange(mean.life)
# A tibble: 5 x 2
  continent mean.life
  <fct>         <dbl>
1 Africa         48.9
2 Asia           60.1
3 Americas       64.7
4 Europe         71.9
5 Oceania        74.3
```

51

51

Summarising Continuous Data

Note: you need a reasonably large sample for reliable information about the shape of a population distribution



52

52

Summarising Continuous Data: Shape

- Graphical summary: Boxplot, Histogram

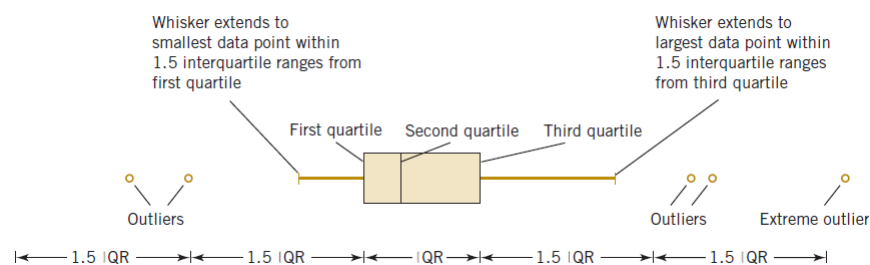
53

53

Boxplot

- A boxplot is a graphical display showing **centre**, **spread**, **shape**, and **outliers**.
- It displays the **5-number summary**:

min, Q_1 , median, Q_3 , and max



54

54

Boxplot of Breaking Strength

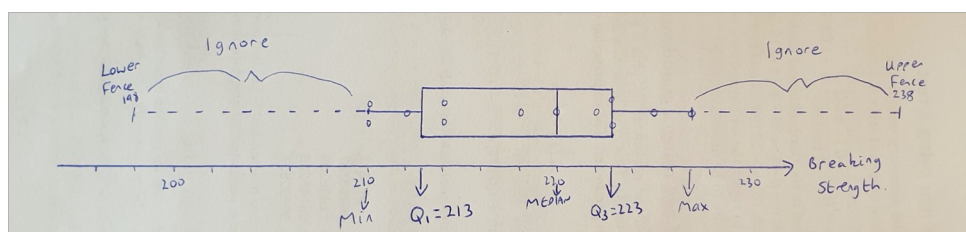
Data: breaking strength of wire in kilograms

220 214 222 218 223 210 223 210 227 225 212

Variable	Minimum	Q1	Median	Q3	Maximum
Breaking Length	210.00	213.00	220.00	223.00	227.00

Upper fence: $Q_3 + 1.5 \text{ IQR} = 223 + 1.5 \times 10 = 238$

Lower fence: $Q_1 - 1.5 \text{ IQR} = 213 - 1.5 \times 10 = 198$

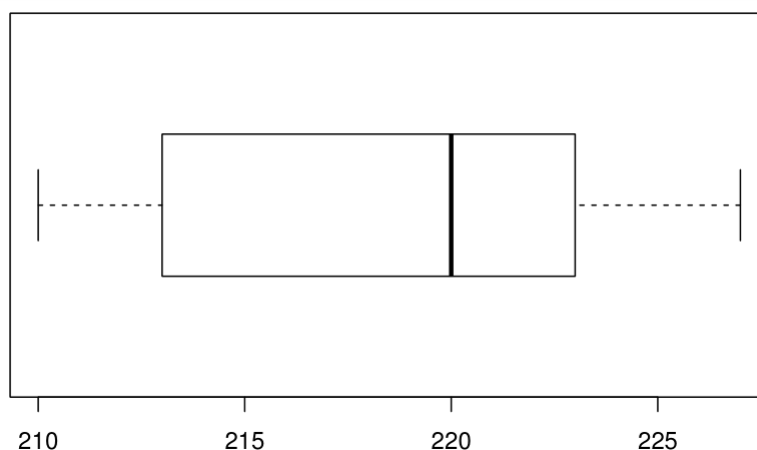


55

55

Graphical Summary in R: Vector

```
boxplot(wire.strength, horizontal=TRUE)
```



Note: `boxplot` gives slightly different result, as it uses a different method to calculate the quartiles (and fences)
Usually these differences are minor so can be ignored, as in this case

56

56

Histograms

- ✓ Useful to show general shape, location and spread of data values – representation by **area**

Construction

- Determine range of data – *minimum, maximum*
- Split into convenient intervals (or bins)
- Usually use 5 to 15 intervals
- Count number of observations in each interval - *frequency*

57

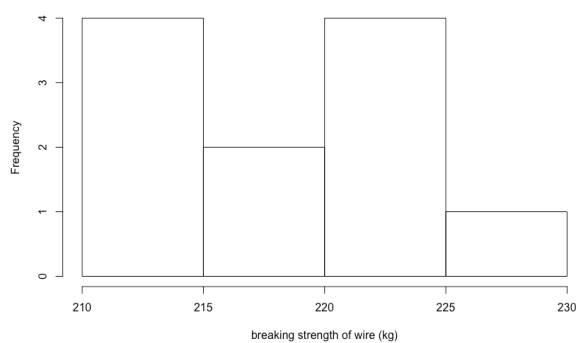
57

Histogram of Breaking Strength

Data: breaking strength of wire in kilograms

220 214 222 218 223 210 223 210 227 225 212

- **Find the minimum and maximum**
- **Make classes of width 5 starting from minimum**
- **Count the frequency**
- **Plot the histogram!**



59

59

Shape of the histogram

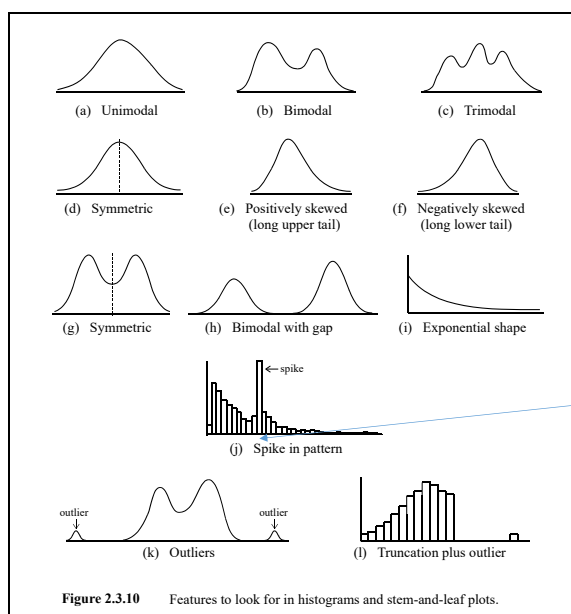
When talking about the shape of the histogram, make sure to address the following three questions:

1. Does the histogram have a single, central hump or several well separated bumps?
2. Is the histogram (or boxplot) symmetric? Or more spread out in one direction, i.e. skewed
3. Any unusual features? e.g. outliers, spikes

60

60

Features to look for



e.g. minimum value for free postage!!!

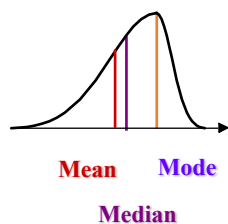
61

61

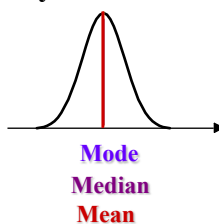
Remember the mean, median and mode ?

The mean is the average data value,

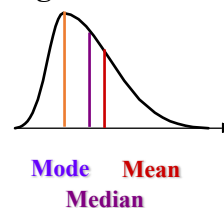
Left-Skewed



Symmetric



Right-Skewed



The value of the mean is strongly affected by skewness and outliers, - more so than the median.

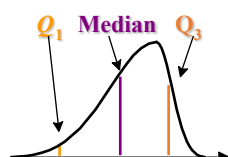
62

62

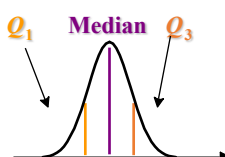
Shape & Box Plot

These shapes can also be seen in the boxplots

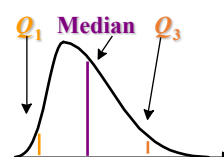
Left-Skewed



Symmetric



Right-Skewed



Left skewed - Longer tail on left than right, median may not be central in the box.

63

63

The mean-median-mode rule of thumb does not always apply

There are exceptions to this rule, eg where the data are **multimodal** or there are **unusual outliers**, so best to look at a histogram too.

For a **small sample**, the boxplot is often very asymmetric even though the underlying distribution is symmetric (see example R code).

For **small samples, you cannot use the boxplot to determine if a population distribution is symmetric**. Histograms are more reliable for small samples, but it can still be difficult if the sample is too small to provide reliable information about the shape of the distribution.

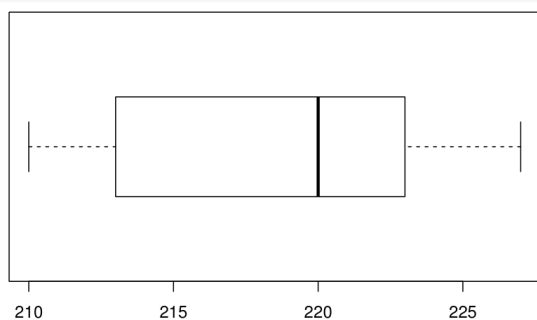
If the dataset is small, boxplots don't provide reliable information about the shape of the distribution

64

64

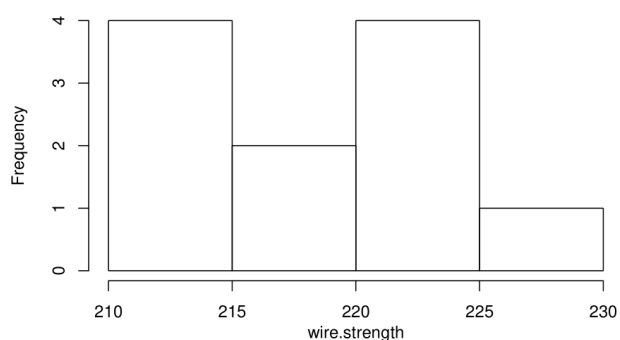
Graphical Summary in R: Vector

```
boxplot(wire.strength, horizontal=TRUE)
```



```
hist(wire.strength)
```

Histogram of wire.strength



65

65

Graphical Summary in R: Dataframe

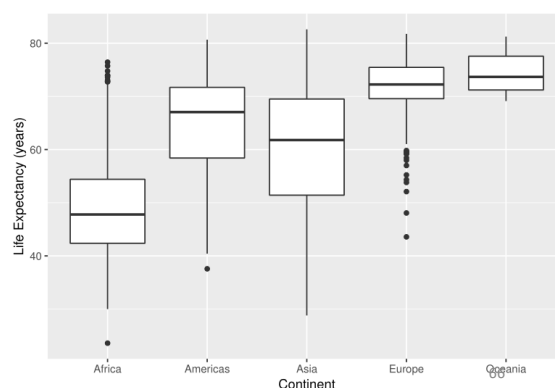
Plot the **boxplot** of life expectancy for gapminder data:

```
ggplot(gapminder, aes(y = lifeExp)) +  
  geom_boxplot() +  
  ylab("Life Expectancy (years)")
```



Plot the **boxplot** of life expectancy for different continents:

```
ggplot(gapminder, aes(x = continent, y = lifeExp)) +  
  geom_boxplot() +  
  ylab("Life Expectancy (years)") +  
  xlab("Continent")
```



66

Explanatory and response variables

TIP: Explanatory and response variables

To identify the explanatory variable in a pair of variables, identify which of the two is suspected of affecting the other and plan an appropriate analysis.

explanatory variable $\xrightarrow{\text{might affect}}$ response variable

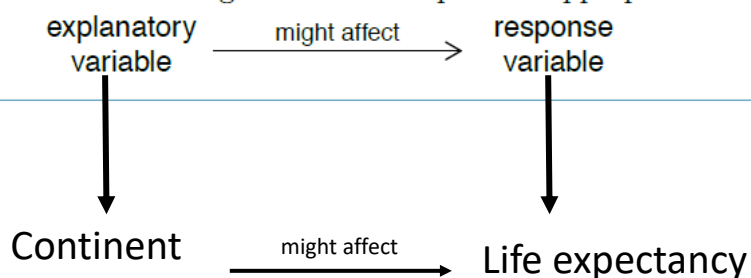
67

67

Explanatory and response variables

TIP: Explanatory and response variables

To identify the explanatory variable in a pair of variables, identify which of the two is suspected of affecting the other and plan an appropriate analysis.



68

68

EDA: graphical summaries of data

- Depends on the variable of interest
- **Categorical** response variable: **barchart** (n or %) or pie chart (not recommended)
- **Categorical** response variable **with a categorical explanatory variable**: **grouped barchart**
- **Continuous** response variable: **histogram**, **boxplot**, density plot
- **Continuous** response variable with **a categorical explanatory variable**: **grouped boxplot**

69

69